

An aligned bit circuit with cheaper centers and a conditional integer-multiplication saving $1624/10^{12}$

eumemic*

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Abstract

With fast Gaussian resampling the conditional witness is limited by the bit network: $\kappa < a_b/2$. We make two changes to the paired bit network at $h = 50$. The side circuit aligns its blocks across common points and shares each top-level pair-star chain between the two groups that use it: 494196 roles per invocation, including the center wires, instead of 509244. Each center wire now takes its value from the group total computed inside the side circuit, so its label rises only to the star span H_i , of dimension $h - 1$, before the scatter; it loses $h - 1$ instead of h dimensions per invocation in both stage directions. The bit saving rises from $296/10^{11}$ to $325/10^{11}$, and the conditional witness from $1479/10^{12}$ to $\kappa = 1624/10^{12} > 2^{-30}$.

1 Scope

The source is OpenAI’s *Integer multiplication below $n \log n$* , pinned at `adc7f1241b42e322a6451854ab7e4b4c146bf7` with the compact-control construction, the compressed complex network and fast Gaussian resampling. Only the bit network changes. Its construction, the paired network of the compact-control manuscript, is assumed with its notation: the form $I - J/9$ on $F = \mathbb{Q}^{50}$, stage frames $D_0 = B \otimes F$, $D_1 = A \otimes F$, $D_U = (B \otimes F) \perp (P \otimes U)$, X_t , Y_t , the transparent schedule, and the stage-1/stage-3 bank sharing. The original central loss is $P \otimes F$, of dimension h , per center wire and invocation.

2 Construction

2.1 An aligned bit circuit with cheaper centers

We change the bit network of Section 1 in two ways: a smaller side circuit, and center wires that lose $h - 1$ rather than h dimensions per invocation. The ground size $h = 50$, the form $I - J/9$, the stage frames of the data banks, and the bank sharing between stages 1 and 3 are unchanged.

Aligned blocks. In the weighted paired recursion for the common point i , put the points $2k, 2k+1$ in one block whenever both differ from i ; the partner i' of i forms a singleton. At every coarser level group blocks $2k, 2k+1$ in the same way. All groups therefore use the same blocks away from their own point, and the identification of equal pair-star sums across groups applies more often.

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Shared pair-star sums. At the top level of group i , a vertex a whose partner q lies in its block A needs, for each other block J , the sum $Y_i(a, J)$ of the triples $\{i, q, c\}$ with $c \notin A \cup J$. Let $\mathcal{C}$ consist of the blocks other than A and the block of i . Form the prefix and suffix sums, in increasing block order over $\mathcal{C}$, of the pair-star block sums $\sum_{c \in \mathcal{C}} x_{iqc}$. These chains depend only on the unordered pair $\{i, q\}$, since $\mathcal{C}$ consists of the full blocks avoiding both i and q ; group q forms the same chains for its vertex i' . For $J \in \mathcal{C}$, $Y_i(a, J)$ is the prefix-suffix sum plus the single triple $\{i, q, i'\}$; for J the block of i it is the total over $\mathcal{C}$. The rest of the recursion is unchanged.

Totals. Each group also keeps its total $z_i = \sum_{T \ni i} x_T$, the top-level total of its recursion. Its source span is $H_i = \langle t_T : T \ni i \rangle$, which is positive definite by the common-point argument above.

Equal supports are identified across groups as before, by the fixed pair and the third points of a pair-star. At $h = 50$ the retained graph has $c = 435346$ additions, the $q = 58800$ partial outputs, and $h = 50$ totals. Exact support comparison checks every partial output and total, the disjointness of every addition, and the common point of every node.

Centers from the totals. Let the center wire C_i be the role carrying z_i out of the side graph. Compile roles as before, treating the h center uses like designated outputs, so $R = c + q + h = 494196$ roles suffice. Let L' be the product of the gates of this graph, V the source copy, J the injection of the partial outputs, R_0 the original central scatter, and $J' = JR_0$. The forward schedule is

$$L', J', L'^{-1}, V, L', J', L'^{-1}, V^{-1}.$$

For arbitrary scratch z the injections give $J'L'z + J'L'(z + Vx) = J'L'Vx$. The center roles of $L'Vx$ carry $z_i = (Gx)_i$, so $J'L'V = JLV + R_0G = I$ and the invocation is $y \leftarrow y + x$; the cleanup restores every auxiliary value. Stage 2 reverses and inverts this schedule.

In a forward invocation the frames are D_0 for the first three operations, X_{t_X} for V , the graded frames D_{U_z} for the middle L' , D_0 for R_0 , Y_{t_Y} for J , and D_1 for the last two operations. Thus C_i passes through D_{H_i} at its gate in the middle L' and returns to D_0 at R_0 , losing $P \otimes H_i$, of dimension $h - 1$. The physical Y wires pass $B \otimes \langle t \rangle \subset D_0 \subset Y_t$, and the physical X wires pass $A \otimes \langle t \rangle \subset X_t \subset D_1$.

In stage 2 the operations are

$$V, L', J, R_0, L'^{-1}, V^{-1}, L', J, R_0, L'^{-1},$$

with logical source Y and target X . Their frames are D_0, D_0, X_{t_X}, D_1 , the complement frames $D_{U_z^\perp}, Y_{t_Y}$, and then D_1 . The center C_i rises from D_0 to D_1 at the first R_0 and falls to $D_{H_i^\perp}$ at its gate in L'^{-1} , again losing $P \otimes H_i$. Since H_i is positive definite, $H_i^\perp$ is nondegenerate. The physical X wires pass $A \otimes \langle t \rangle \subset X_t \subset D_1$, and the partial-output roles pass $\langle t_X \rangle \subset U_z^\perp$ as before. Every other edge is one of the nested edges already treated. Each auxiliary role starts at D_0 and ends at D_1 , so the stage-1/stage-3 bank sharing applies unchanged.

Proposition 1 (Aligned bit interface). *The aligned bit network restores arbitrary scratch, exchanges the data banks, and satisfies the original all-role endpoint identity with*

$$\begin{aligned} W_b^{\text{al}} &= 2N + 2v^2R = 394759742720000, \\ s_b^{\text{al}} &= W_b^{\text{al}}m - N + 6v^2h(h-1) = 49344965957616000000, \\ W_b^{\text{al}}m - s_b^{\text{al}} &= 1882384000000 > 0. \end{aligned}$$

Proof. The scalar schedule and both frame assignments were given above. The only decreasing edges are the $3v^2h$ center returns, each of dimension $h - 1$. Telescoping gives total absolute label change $W_b^{\text{al}}m - 2N + 6v^2h(h - 1)$, and the unchanged negative source projections add N rational rank. Data endpoints are unchanged and every auxiliary role has endpoints zero and I_m , with its scalar value restored. The common-frame identity and rational-shear transfer apply to these fixed finite data. $\square$

The exact deficit is $\eta_b^{\text{al}} = 49/1284490000$, against $23/661055000$ for the paired network. With $\log m < 11737/1000$, $\eta_b^{\text{al}} > (325/10^{11})(11737/1000)$, so $s_b^{\text{al}}/W_b^{\text{al}} < m^{1-325/10^{11}}$.

3 Assembly

The fast-Gaussian assembly is unchanged except for τ : $a = 325/10^{11}$, $a_c = 14/10^9$, $\epsilon = 49999/100000$, $c = 9999/10000$, $\beta = 19/25$, $\zeta = 1/10^4$, $\delta = 1/10^6$, $C_1 = 19601/10000$, $\lambda = 1 - 32495/10^{13}$, $\lambda' = 1 - 3249/10^{12}$. The leaf needs $(1 - \beta)a_c = 3.36 \cdot 10^{-9} > 1 - \lambda'$, and the guard needs $\beta > 3/4$; both hold. The minimum margin is $g_3 = \epsilon(1 - \lambda') = 162446751/10^{17} > 1624/10^{12} = \kappa$, so

$$T(n) = O(n(\log n)^{1-1624/10^{12}}), \quad 2^{-30} < \frac{1624}{10^{12}} < 2^{-29},$$

conditional on the identified upstream and extension proofs. The scoped ceiling for this bit network is $\kappa < a/2 < 2^{-29}$. A larger bit saving would also need $(1 - \beta)a_c > a$ with $\beta > 3/4$, so a_c or the guard would have to improve too.

4 Reproducibility

`scripts/bit_circuit.py` builds the circuit and checks every partial output, total, addition and compiled role frame in both directions; `scripts/aligned_bit_network.py` records the counts and witness in `certificates/aligned-bit-network.json`. `tests/test_aligned_bit_network.py` simulates the new schedule exactly over $\mathbb{F}_2$ with dirty scratch in both stage directions for $h = 8, \dots, 14$, and checks the star span, the counts, the parameter windows and the patch text. These checks support the written argument; they are not formal verification, and the construction has not been independently reviewed.